

Chapter 17 — Principles of Urologic Endoscopy

- Endoscopy has reshaped urology more than most disciplines: **bladder outlet obstruction, urothelial tumours, ureteric obstruction and nephrolithiasis** are all now diagnosed and treated endoscopically.

HISTORY OF ENDOSCOPY

Rigid and Optical Line

- **1806, Philipp Bozzini — the Lichtleiter (“Light Conductor”), the first “modern” endoscope.** Angled mirrors conducted **candlelight** from a sharkskin-covered box down aluminium tubing into the body.
- **1853, Antonin Jean Desormeaux (French urologist) — smaller-profile endoscope with improved mirrors and a kerosene lamp.** Through it he excised a urethral papilloma, making him the **first to perform a therapeutic endoscopic procedure.**
- **1877, Max Nitze — two conceptual leaps at once.** He moved illumination (a **water-cooled electric platinum filament**) to the **tip** of the instrument, and used a **series of optical lenses at precise distances** along a hollow air-filled scope to relay and magnify the image.
- **1906 — the Amici prism,** allowing 90-degree offset images to retain correct orientation. Essentially the only design advance between Nitze and Hopkins.
- **1966, Harold Hopkins — the rod-lens system.** Hollow-chamber relay lenses were abandoned for **glass rods with short air gaps, which increased light transmission while decreasing scope calibre.**
- **Rod-lens optics underpin all contemporary rigid cystoscopes** and made possible the **first rigid ureteroscope** in 1979.

Flexible and Fibreoptic Line

- **1854, John Tyndall — light travels along a curved stream of water by internal reflection.** This is the physical principle, not an instrument.
- That principle led to drawing **molten glass into flexible small-calibre fibres** for light transmission; **fibre optics made flexible endoscopes possible.**
- **1964, Marshall — first reported flexible ureteroscopy (URS).**
- **Flexible URS remained of limited utility until a working channel and active tip deflection were added in the early 1980s.** Both features are required before a flexible scope is therapeutically useful.
- **Fibreoptic endoscopes remain widely used but are fragile with limited optical resolution — the two drawbacks that drove digital conversion.**

Digital and Single-Use Era

- **1970, Boyle and Smith — the charge-coupled device (CCD),** a sensor converting photons into a digital image. Over three decades, miniaturisation allowed the **CCD to be mounted on the distal tip of the endoscope.**
- **2005, ACMI DCN-2010 flexible cystoscope — first commercially available digital endoscope.**
- **2006, Olympus Invisio DUR-D — first digital ureteroscope.**
- **Digital (“chip-on-tip”) sensors give better image resolution and durability than rod-lens or fibreoptic bundles, and dispense with a separate light cable and camera.**
- **2011, PolyScope (Lumenis) — the first single-use ureteroscope,** driven by the **damage rate and reprocessing costs of reusable fibreoptic ureteroscopes.** It used a **reusable fibreoptic bundle attached to disposable flexible catheters; limited deflection gave poor success for lower pole stones and it was never widely adopted.**

- **January 2016, LithoVue (Boston Scientific) — the first digital single-use ureteroscope**, marking the start of the current era.

BASIC EQUIPMENT AND VIDEO-ENDOSCOPIC SYSTEMS

Irrigation Fluid

- Cystourethroscopy requires three things: **irrigation fluid, a light source, and an endoscope.**
- **Normal saline is the default irrigant.**
- **Where monopolar electrocautery is used, the irrigant must be electrolyte-free — glycine or mannitol,** since saline would disperse the current.
- **Bipolar electrocautery permits normal saline for transurethral procedures, thereby avoiding the complications of the electrolyte-free solutions.**
- **Sterile water is to be avoided — risk of absorption, electrolyte disturbance and haemolysis.**

Light Source and Enhanced Imaging

- **A high-intensity xenon or halogen external light source** delivers white light via a fibreoptic cable and the endoscope to the field of view.
- Modern units incorporate **light level detection with automatic output adjustment** for constant illumination.
- **Narrow-band imaging (NBI)**, introduced into urology in 2007, uses only blue (415 nm) and green (540 nm) wavelengths. These are strongly absorbed by haemoglobin, so urothelial capillaries, small papillary lesions and carcinoma in situ stand out. No drug instillation is required.
- **Blue-light cystoscopy — approved by the U.S. Food and Drug Administration (FDA) in 2010.** Sequence: intravesical instillation of hexaminolevulinate one hour before cystoscopy → the molecule accumulates in the mitochondria of urothelial tumour cells but is eliminated from normal urothelium → the bladder is illuminated with blue light at 360–450 nm → porphyrins in abnormal cells emit red fluorescence while normal cells appear blue.

Video-Endoscopic Systems

- Direct viewing through the **optical eyepiece** remains possible, but a **camera displaying onto a monitor on a fixed or mobile video-endoscopic tower** is standard (Fig. 17.1).
- **Advantages of video-endoscopic units:** better visualisation, aiding patient safety and surgical training; reduced bodily fluid exposure; and improved surgeon ergonomics.
- **Traditional system — four components:** light source, endoscope camera, image processor/recorder, monitor.
- **Newer digital endoscopes eliminate the external camera and integrate the light source and video cable into a single built-in housing** (Fig. 17.2).

Signal Pathway

- **Traditional system:** rod-lens or fibreoptic bundle relays the optical image to the eyepiece → **external camera CCD chip generates an analogue voltage waveform → video processor converts it into a digital video signal for the monitor.**
- **Digital endoscope system:** the **CCD chip sits at the distal tip.** The image is converted to an electrical signal immediately, managed by the video processor **without internal optics or an external CCD camera.**

CYSTOURETHROSCOPY

Indications

- **Cystourethroscopy is the commonest office-based procedure in urology.** Direct visual examination of bladder and urethra; performed routinely in both office and operating room.
- Predominantly **diagnostic**, but permits **transurethral therapeutic procedures**. Indications are tabulated in Table 17.1.
- **Evaluation of microscopic or gross haematuria is among the commonest indications.**
- Allows **collection of cytological specimens** and **retrograde pyelography in patients who cannot receive intravenous contrast.**
- **Surveillance of urothelial carcinoma** is the other common indication.
 - **In-office biopsy and fulguration** of small urothelial lesions is feasible.
 - **Upper tract surveillance** via **selective ureteric catheterisation with retrograde pyelography**, **upper tract washings for cytology**, and **brush biopsy for histology.**
- **Lower urinary tract signs and symptoms:** recurrent infection, voiding dysfunction, chronic pelvic pain.
- **Treatable in-office in selected cases: urethral strictures, bladder stones, foreign bodies.**
- **With fluoroscopy, ureteric stents can be placed or exchanged in the clinic.**

Equipment

Sizing Convention

- Cystourethrosopes come **rigid and flexible, in a range of sizes**; dimensions are expressed in **French (Fr) gauge**.
- **French gauge devised by Joseph-Frédéric-Benoît Charrière (1803–1876), French instrument designer. 1 Fr = 1/3 millimetre in diameter, equivalently 3 Fr = 1 mm.**

Rigid versus Flexible

- **Rigid — optics:** uses the **Hopkins rod-lens system**, giving **better optical clarity** than the fibreoptic bundles of flexible scopes.
- **Flexible — optics:** fibreoptic bundle or digital chip; optically inferior to rod-lens.
- **Rigid — working channel:** larger channels → **greater irrigation flow and a wider range of usable instruments.**
- **Flexible — working channel:** single, smaller channel shared between irrigation and instruments.
- **Rigid — handling:** easier one-handed control, freeing the second hand for ancillary tools.
- **Flexible — handling:** active tip deflection eases **complete bladder inspection and negotiation of a high bladder neck or prostatic median lobe.**
- **Rigid — positioning:** requires **frog-leg or lithotomy position.**
- **Flexible — positioning:** no frog-leg or lithotomy position required; smaller calibre improves patient comfort → the instrument of choice for office-based procedures.

Rigid Cystourethrosopes

- Manufactured as a **set: optical lens, bridge, sheath, obturator** (Fig. 17.3).
- **Optical lens tip angles range from 0 to 120 degrees:**
 - **0 or 12 degrees — urethra**, the best view for urethral visualisation.
 - **25 or 30 degrees — therapeutic work.**
 - **70 or 120 degrees — anterior and inferolateral bladder walls, dome and bladder neck.**

- **Bridge connects the optical lens to the sheath.**
 - **Diagnostic bridge** — no working channel.
 - **Therapeutic bridge** — one or two working channels.
 - **Albarran's bridge** — contains a lever that deflects wires and catheters emerging from the working channel, to facilitate cannulation of the ureteric orifice (Fig. 17.4). Indicated in elevated bladder neck, large prostatic median lobe, or ureteroneocystostomy.
- **Sheaths** are marked with **sheath size and associated working channel size** (Fig. 17.5).
 - **15 and 17 Fr sheaths** — diagnostic cystoscopy.
 - **Larger sheaths** — therapeutic procedures, where improved irrigant flow and larger working channels are needed.
- **Obturator** blunts the distal end of the sheath to allow **passage into the bladder without visual guidance.**

Flexible Cystourethroscopes

Specifications

- **Diameter 14–17 Fr.**
- **Working length 37–40 cm** in most models.
- **“Inline” eyepiece lens**; field of view approximately 120 degrees.
- **Tip deflection 120–220 degrees**, and is either:
 - **Intuitive** — tip deflects in the same direction as the lever, or
 - **Counterintuitive** — tip deflects opposite to the lever.
- **Single working channel, 6.4–7.5 Fr**, shared by irrigation and instruments.
- Models differ in **tip deflection, direction of view, field of view, working channel size, illumination and optics.**
- **Photodynamic and narrow-band imaging capability** increasingly common.
- Available in **fibreoptic and digital** versions; **digital scopes** require neither focusing nor white balancing.

Fibreoptic versus Digital: In Vitro Comparison

- An in vitro study compared **resolution, contrast evaluation, depth of field, colour representation and illumination** across **fibreoptic, standard-definition and high-definition** flexible cystoscopes, all from a single manufacturer.
- **High-definition scope**: markedly superior resolution and depth of field, and a substantially larger image than standard-definition.
- **Colour representation** only marginally better; no difference in contrast evaluation between the three.
- **Illumination** was significantly better in the fibreoptic scope than in either digital model — the counterintuitive result worth carrying.

Fibreoptic versus Digital: Clinical Durability

- A large randomised comparison of **fibreoptic versus standard-definition digital** flexible cystoscopes found only a **non-significant trend favouring the digital scopes on surgeon optical ranking**, and **no difference in durability.**
- **Repairs were rare**, and both damaged scopes were injured on placement into a storage case rather than during use — the storage step, not the procedure, is the vulnerability.
- Independent repair-company data: **flexible cystoscopes require fewer than one repair every two years; the commonest repair is to the rubber overlying the distal flexible segment.**

- **Unlike flexible ureteroscopes, flexible cystourethrosopes are robust** — so the **optical mechanism is a lesser determinant of durability** in cystoscopes.

Single-Use Cystoscopes

- **Ambu (Ballerup, Denmark) launched the world's first single-use endoscope in 2009, and the first disposable cystoscope — the aScope 4 Cysto — in 2020.**
- Subsequent single-use cystoscopes: **WiScope Cysto (innoMedicus), Uro-G (UroViu), NeoFlex (NeoScope).**
- **Benchtop comparison of a single-use scope against a reusable Olympus model showed comparable manoeuvrability, image quality and illumination.**
- **In vivo studies comparing clinical outcomes of reusable versus disposable cystoscopes are lacking** — the current evidence gap.
- **Environmental and economic assessments favour single-use: reduced water usage, comparable costs, and a significantly lower carbon footprint** than reusable scopes.

Patient Preparation

- **Informed consent is mandatory before any cystoscopic procedure.**
- **Urinalysis and urine culture where indicated; all urinary tract infections must be treated before instrumentation, because of the risk of bacteraemia and sepsis following lower urinary tract manipulation.**

Antimicrobial Prophylaxis

- Two guideline sources:
 - **American Urological Association (AUA) Best Practice Policy Statement on Urologic Surgery Antimicrobial Prophylaxis** — published 2008, most recently updated 2019.
 - **European Association of Urology (EAU) Urological Infections guidelines** — first published 2001, updated through 2023.
- **Neither body recommends prophylaxis for routine diagnostic cystoscopy in the absence of patient-related risk factors** (Table 17.2).
- **AUA — therapeutic lower urinary tract procedures, meaning transurethral resection of bladder and of prostate: prophylaxis for less than 24 hours.** The basis is a meta-analysis of prophylaxis before transurethral resection of the prostate (TURP), which showed **reductions in bacteriuria, bacteraemia, symptomatic urinary tract infection and high-grade fever.**
- **EAU — routine prophylaxis for TURP; for transurethral resection of bladder tumour, prophylaxis only in patients at high risk of postoperative sepsis.** This is the **examinable divergence between the two bodies.**
- **AUA agents of choice for transurethral procedures: cefazolin or trimethoprim-sulfamethoxazole.**
- **AUA alternatives: an aminoglycoside with or without ampicillin; another first- or second-generation cephalosporin; amoxicillin/clavulanate.**
- **Against rising regional resistance, the EAU position is that knowledge of local pathogen profiles, susceptibility and virulence is mandatory** — the local antibiogram overrides a fixed agent list.

TABLE 17.2 PATIENT RISK FACTORS PRECLUDING SIMPLE CYSTOSCOPY WITHOUT ANTIMICROBIAL PROPHYLAXIS

AUA Best Practice Policy Statement	EAU Urological Infections Guidelines
Advanced age	Older age
Poor nutritional status	Malnutrition
Immunodeficiency	Impaired immune system

AUA Best Practice Policy Statement	EAU Urological Infections Guidelines
Externalised catheters; colonised endogenous or exogenous material	Indwelling catheters and bacterial burden
Anatomical anomalies of the urinary tract	Previous instrumentation
Smoking	Diabetes mellitus
Chronic corticosteroid use	Extremes of weight
Distant coexistent infection	History of urinary tract infection
Prolonged hospitalisation	Genetic factors

- **Shared across both lists:** age, malnutrition, immune compromise, catheters or colonised material.
- **AUA distinctive:** smoking, chronic corticosteroid use, distant coexistent infection, prolonged hospitalisation, anatomical anomalies.
- **EAU distinctive:** diabetes mellitus, extremes of weight, prior urinary tract infection, previous instrumentation, genetic factors.
- **What the table tests:** not the individual items but the principle that **prophylaxis is patient-driven, not procedure-driven, for diagnostic cystoscopy**. The trap it sets is treating the two columns as contradictory when they are two roads to the same rule.

Skin Antisepsis

- Commercial agents contain **iodophors or chlorhexidine gluconate**, in **aqueous or alcohol-based** solution.
- **Chlorhexidine gluconate and alcohol-based solutions damage mucous membranes and are therefore not recommended for the genitalia.**
- **Aqueous iodophor-containing products (povidone-iodine) are safe on all skin surfaces regardless of patient age** — the default for genital preparation.

Urethral Lubrication and Anaesthesia

- After antisepsis, a **lubricating gel is instilled into the urethra** for flexible cystourethroscopy; **either plain or lidocaine gel**.
- **In males, meta-analysis of randomised trials shows lidocaine gel significantly reduces pain.**
- **Dwell time is the determinant of efficacy: it must exceed 5 minutes, ideally 10 to 20 minutes.** Instilling and proceeding immediately negates the benefit.
- **In females the evidence is equivocal** — one randomised trial showed benefit, another showed no pain reduction.

Non-Pharmacological Comfort Measures

- **In most males the least comfortable part of flexible cystourethroscopy is passage through the membranous urethra.** Every measure below targets this moment.
- **“Bag squeeze” — manual compression of the irrigation bag to raise hydrostatic pressure during passage through the membranous urethra** significantly reduced pain compared with gravity irrigation alone in a double-blind randomised trial.
- **Instructing the patient to urinate, thereby relaxing the external urethral sphincter**, likewise improves pain during flexible cystoscopy.
- **Allowing the patient to watch the procedure on the video monitor:** randomised evidence in **males** shows **less pain and lower post-procedure pulse rates**; in **females undergoing office cystoscopy with a 17 Fr rigid scope** there was **no difference in procedural pain**. The divergence between the sexes is the point.

- **Music:** males randomised to **classical music** during flexible cystoscopy had **less pain, higher satisfaction, and lower post-procedure pulse rates and systolic blood pressures**; a subsequent meta-analysis confirmed reduced pain and anxiety.

Technique

Preparation and Meatal Access

- Inspect the external genitalia first for cutaneous lesions and anatomical abnormalities, before any instrument is passed.
- Mild meatal stenosis may be managed with sequential metal dilators.
- Dilate to at least 2 Fr wider than the intended endoscope size — the working rule.

Insertion in Females

- Rigid cystourethroscope insertion in females is safest using the sheath obturator.
- The scope must often be directed anteriorly as it advances into the bladder.
- Flexible cystourethrosopes are usually inserted like a Foley catheter, with active deflection used as required.

Insertion in Males

- The penis is placed on maximal stretch to straighten the urethra.
- Rigid scope: the penis is grasped with all five digits of the non-dominant hand.
- Flexible scope: the penis is pinched between the fourth and fifth digits of the non-dominant hand at the corona; the thumb and index finger advance and direct the scope (Fig. 17.6).
- Morbidly obese patient: lay the scope down, retract the pannus with the non-dominant hand, and direct the flexible scope tip into the urethra like a Foley catheter; once in the mid-penile urethra, transfer to the dominant hand and advance normally.

Passage Through the Urethra

- Angle the penis 45 to 90 degrees relative to the abdominal wall while traversing the anterior urethra.
- Beyond the membranous urethra, direct the cystoscope anteriorly to enter the bladder:
 - Flexible — active upward deflection.
 - Rigid — drop the distal (eyepiece) end of the scope toward the operating table.
- Advance under maximal irrigation, evaluating the lower urinary tract systematically.
- Inspect the penile and bulbar urethra for strictures.
- The periurethral glands of Littre drain into the urethra dorsally — note them.
- Encourage relaxation, particularly in young males, during passage through the membranous urethra, the least comfortable segment.
- Prostatic urethra: identify the verumontanum and utricle posteriorly; measure the length of the posterior urethra and assess the size of the prostatic lobes.

Systematic Bladder Inspection

- Begin rigid cystoscopy with a 25- or 30-degree lens.
- Survey the bladder floor and trigone.
- Note the number, location and configuration of the ureteric orifices, and observe efflux from each ureter for gross blood.
- Inspect the remainder of the bladder for: stones, trabeculation, cellules, diverticula, erythematous patches, and papillary and sessile lesions.
- Lateral walls — rotate the cystoscope while keeping camera orientation fixed.

- **Dome, anterior and posterolateral walls — change to a 70- or 120-degree lens.**
- **At completion: empty the bladder, then withdraw the endoscope.**
- **Exception: if a Foley catheter is to be placed afterwards, leave the bladder at least partially full before removing the cystoscope.**

Special Circumstances

Suprapubic Cystostomy

- **Indications for cystoscopy are the same as in patients without a chronic indwelling catheter.**
- **These patients carry increased risk of infection, bladder calculi and bladder cancer.**
- **However, no level 1 data show that surveillance cystoscopy improves survival in long-term indwelling catheter patients.**
- **The 2021 AUA/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) Guideline on adult neurogenic lower urinary tract dysfunction states that screening or surveillance cystoscopy should NOT be performed in patients with chronic indwelling catheters. Elevated cancer risk does not translate into a surveillance recommendation.**
- **Route: transurethral or via the suprapubic tract. Many long-term suprapubic tube patients have urethral stricture disease, making the suprapubic tract the only feasible route.**
- **Avoid endoscopy through an immature suprapubic tract; maturation usually takes several weeks from creation.**
- **When an immature tract must be used: place a wire through the suprapubic tube before removing it. This safety wire maintains bladder access and guides scope placement; it is also valuable in morbidly obese patients with long, tortuous tracts.**
- **Ureteric access through a cystostomy tract with a rigid scope is difficult because of the acute angle required to cannulate the orifice. Solutions, in escalating order:**
 - **Angled ureteric catheters or an Albarran deflector.**
 - **A flexible cystoscope, which usually overcomes the angle problem.**
 - **In low-grade urethral stricture, a semirigid ureteroscope passed transurethrally, restoring standard ureteric access.**
- **Ureteric orifices may be obscured by oedema from the chronic suprapubic tube — give intravenous indigo carmine or methylene blue early in the procedure to visualise them.**

Continent Urinary Diversions

- **Two general classes after cystectomy:**
 - **Orthotopic diversion — anastomosed to the urethra; continence depends on the striated urinary sphincter.**
 - **Continent cutaneous reservoir — a catheterisable channel from the reservoir to the anterior abdominal wall.**
- **The catheterisable channel may be: appendix (Mitrofanoff), tapered or imbricated terminal ileum with ileocaecal valve, or an intussuscepted nipple valve.**
- **Obtain the operative note before any endoscopic procedure on a continent diversion. Four things must be known: bowel segment used; type and location of the ureteroenteric anastomoses; continence mechanism employed; whether an afferent limb was created.**

- **Orthotopic diversion** — transurethral access is usually straightforward with a rigid cystoscope.
 - If a urethral anastomotic contracture is present but the patient is not obstructed, use the smallest possible scope rather than dilating or incising the stricture — dilation or incision risks urinary incontinence.
- **Continent cutaneous reservoir** — diagnostic endoscopy is best performed with a flexible endoscope through the catheterisable channel.
 - All but the most straightforward therapeutic procedures should be done percutaneously, because continence mechanisms are fragile; excess manipulation causes stomal stenosis or incontinence.
 - Use preoperative computed tomography or intraoperative ultrasound before percutaneous access, to avoid bowel injury or injury to the reservoir's vascular pedicle.
- **Visualisation within a diversion** is impaired by mucus, mucosal folds, bowel peristalsis and a tortuous afferent limb where present.
- **Irrigation strategy** — the balance is the examinable point:
 - First, irrigate out all mucus.
 - Then irrigate judiciously. Too little irrigation makes mucosal folds more prominent and impairs visualisation.
 - Overdistension prevents access to the afferent limb and causes troublesome retroflexion of flexible cystoscopes.
- Angled catheters and indigo carmine or methylene blue may again be required.

KEY POINTS: CYSTOURETHROSCOPY

- Rigid cystourethrosopes are manufactured in sets: optical lens, bridge, sheath, obturator.
- Flexible cystourethrosopes come in fiberoptic, standard-definition and high-definition digital models.
- Sizes are expressed in French gauge — 3 Fr = 1 mm diameter.
- Select the smallest-diameter cystourethroscope capable of performing the procedure, to reduce genitourinary tract trauma.
- Antimicrobial prophylaxis is not recommended for diagnostic cystourethrosopy unless patient-related risk factors exist.
- Increased hydrostatic pressure during scope passage, and allowing males to observe the procedure, have prospectively been shown to improve comfort during flexible cystourethrosopy.

UPPER TRACT ENDOSCOPY

Indications

Urolithiasis

- Surgical treatment of nephrolithiasis is the commonest indication for upper tract endoscopy.
- Ureteric calculi — AUA/Endourological Society guidelines: either shock wave lithotripsy (SWL) or ureteroscopy (URS).
- Counselling point mandated by the guideline: URS gives a higher stone-free rate but greater morbidity than SWL.
- EAU urolithiasis guidelines: both URS and SWL are appropriate for smaller ureteric stones, but URS is preferred for calculi larger than 10 mm because of its higher stone-free rate.

- **Intrarenal calculi, non-lower pole (AUA/Endourological Society):**
 - Less than 20 mm — SWL or URS.
 - Greater than 20 mm — percutaneous nephrolithotomy (PCNL).
 - **Exception:** patients whose comorbidities make percutaneous treatment prohibitively dangerous have been successfully managed with URS, with high stone-free rates reported for ureteroscopic treatment of stones larger than 2 cm.
- **Intrarenal calculi, lower pole:**
 - Less than 10 mm — SWL or URS.
 - Greater than 10 mm — consider PCNL.

Lower Pole Stones and Shock Wave Lithotripsy

- The EAU guidelines broadly agree with AUA/Endourological Society on intrarenal stone management, with one qualification.
- SWL may be considered for lower pole stones between 10 and 20 mm, provided no unfavourable factors are present.
- **Unfavourable factors that argue against SWL for a lower pole stone:**
 - Shock wave-resistant stone composition: calcium oxalate monohydrate, brushite, or cystine.
 - Steep infundibulopelvic angle.
 - Long lower pole calyx (>10 mm).
 - Narrow infundibulum (<5 mm).
 - Skin-to-stone distance exceeding 10 cm.

Upper Urinary Tract Urothelial Carcinoma

- EAU guidance: perform diagnostic ureteroscopy when imaging is equivocal, or when conservative kidney-sparing management is being considered.
- Ureteroscopic biopsy yields only a small amount of material, yet accurately predicts final pathological grade in the large majority of cases, approximately nine in ten.
- Diagnostic ureteroscopy has not been shown to worsen long-term or disease-specific survival in patients proceeding to extirpative surgery.
- However, it has been associated with an increased risk of intravesical recurrence — hence the requirement for long-term bladder surveillance.
- **Historical position:** endoscopic treatment of upper tract urothelial tumours was reserved for **imperative** indications — solitary kidney, renal insufficiency, bilateral upper tract urothelial carcinoma, or significant medical comorbidity.
- **Current position:** with improved instrumentation and technique, endoscopic treatment of low-grade disease is a reasonable option without compromising survival — elective, not merely imperative, indications now exist.
- Larger tumours are difficult to ablate completely in a single session, prompting reports of percutaneous nephroscopy with electroresection.
- Where feasible, ureteroscopy is preferred over the percutaneous route, because percutaneous access carries a small but reported risk of tumour seeding of the tract and retroperitoneum.

Ureteropelvic Junction Obstruction and Ureteral Stricture

Endopyelotomy

- Despite widespread adoption of laparoscopic and robot-assisted pyeloplasty, endopyelotomy remains a common treatment for ureteropelvic junction (UPJ) obstruction.
- Endopyelotomy may be performed percutaneously or ureteroscopically.
- Advantages of the ureteroscopic approach: it avoids the morbidity of percutaneous renal access and permits an outpatient procedure.
- Two situations where ureteroscopic endopyelotomy is not preferred:
 - Concomitant renal calculi → treat percutaneously, allowing simultaneous stone removal and endopyelotomy.
 - Crossing vessels at the UPJ → decreased endopyelotomy success rate.
- Practical rule: limit ureteroscopic endopyelotomy to patients with documented absence of crossing vessels; treat those with crossing vessels by laparoscopic ureteropyeloplasty.

Endoureterotomy

- Ureteric strictures can be managed ureteroscopically; the technique is nearly identical to ureteroscopic endopyelotomy.
- Endoureterotomy will be less successful in:
 - Strictures longer than 1.5 cm
 - Allograft ureters
 - Previously irradiated ureters
 - Ureters draining poorly functioning kidneys
 - Ureteroenteric anastomotic strictures
- Ureteroscopic incision of short strictures in otherwise healthy ureters is a reasonably successful option.
- Balloon dilation of short strictures may be performed with or without concomitant incision, with moderate durability at one year, roughly half of patients.

Other Indications for Ureteroscopy

- Diagnostic ureteroscopy is indicated for:
 - Persistent, unexplained positive cytology
 - Filling defects on excretory urography
 - Recurrent urinary tract infection localised to a single renal unit
- Miniaturisation of flexible ureteroscopes has substantially improved safety, so diagnostic ureteroscopy can now be performed readily rather than relying on ureteropyelography alone.
- Foreign body retrieval: sutures, proximally migrated ureteric stents, balloon catheters, and fractured working instruments.

Benign Essential Haematuria

- Definition: unilateral gross haematuria with no radiographically defined cause.
- Typically presents after a negative workup — excretory urography, renal ultrasonography, arteriography, or a combination.
- Flexible ureteroscopic inspection of the involved kidney usually achieves both diagnosis and successful treatment.
- Commonest finding: a small haemangioma, which can often be fulgurated.
- Other findings: small venous ruptures, papillary tumours, varices, and calculi.

Equipment

Semirigid Ureteroscopes

Construction and Optics

- Larger rod-lens rigid ureteroscopes persist in some theatres, but smaller-diameter fiberoptic ureteroscopes are less traumatic, less often require ureteric dilation, and are equally capable.
- Semirigid ureteroscopes achieve their smaller diameter through the incorporation of fibre optics.
- Fiberoptic bundle anatomy: multiple glass fibres, each coated with a cladding — a further glass layer of different refractive index. Cladding improves internal reflection, light transmission and durability.
- Fibre arrangement determines function — the core contrast:
 - Randomly grouped fibres, as in a light bundle, transmit light for illumination but no image.
 - Coherently arranged fibres coalesce the light from each fibre to transmit an image.
- Small lenses at both ends of the image bundle create a telescope.
- Image magnification, field of view and focusing ability are determined by the number of fibres in the bundle and the type and orientation of the lenses.
- Changing the axis of the distal-tip lens alters the angle of view, improving visibility of working instruments emerging from the working channel.

Design Advances and Working Channels

- Closer packing of more fibres in the image bundle yields improved image transmission, smaller outer diameters, and larger working channels — in both rigid and flexible ureteroscopes.
- The single light bundle has been replaced by two smaller bundles, permitting a more centrally placed working channel and better light distribution across the field of view.
- Most current semirigid ureteroscopes: tip diameter 7 Fr or less; working channel larger than 3 Fr.
- Channel configuration: either one large single channel, or two smaller independent channels.
- Advantages of two separate working channels:
 - Continued irrigation through one open channel while an instrument occupies the other.
 - Passage of a lithotripsy device through one channel to fracture a stone trapped in a basket in the other — difficult with a single channel because the two instruments become entangled.
- For these reasons, semirigid ureteroscopes with two separate working channels are preferred.

Eyepiece Design

- In-line eyepiece — the common configuration, giving more natural insertion and handling of the scope (Fig. 17.7).
- Offset eyepiece — creates a straight working channel, required for less flexible working instruments: rigid biopsy forceps and pneumatic lithotripsy probes (Fig. 17.8).
- The rise of holmium and thulium laser lithotripsy has reduced the need for offset-eyepiece ureteroscopes, since laser fibres are flexible and do not require a straight channel.

Ureteral Resectoscopes

- Historically, some urologists preferred larger ureteral resectoscopes (11.5 Fr) for resection of large distal ureteric tumours and ureteroscopic endopyelotomy.
- Preoperative ureteric stenting was required to permit passage of the larger-calibre resectoscope, particularly up to the ureteropelvic junction.
- Laser energy for tissue ablation and incision now avoids the greater trauma of these resectoscopes.

Flexible Ureteroscopes

Components and Deflection Mechanism

- Three fundamental components: optical system, deflection mechanism, working channel.
- The non-digital optical system consists of flexible fibreoptic image and light bundles; improvements mirror those described for semirigid ureteroscopes.
- Active deflection permits complete manoeuvrability within the intrarenal collecting system.
- Mechanism, as a sequence: manually operated lever → control wires running the length of the scope → passing through movable metal rings → fixed at the distal end → moving the lever pulls the wire through the rings and deflects the tip.
- “Intuitive” deflection means the tip moves in the same direction as the lever — up is up, down is down.
- Modern flexible ureteroscopes deflect both up and down in a single plane.
- The plane of deflection is designated by the reticle — a notch visible within the field of view (Fig. 17.9). The scope must be rotated to align the plane of deflection with the intended target.
- The active deflection mechanism wears out with repeated use, requiring repair or replacement; each generation improves deflection-mechanism durability.

Reaching the Lower Pole

- Current flexible ureteroscopes permit more than 270 degrees of deflection.
- The deflection required to reach the lower pole varies between patients. Measurement of the ureteroinfundibular angle, between the major axis of the ureter and the lower pole infundibulum, gave an average of 140 degrees and a maximum of 175 degrees.
- 270 degrees of active deflection should therefore permit visualisation of the lower pole in most patients.
- Visualisation is not the same as access. Reaching into the lower pole calyx with the tip in a position that allows work to be done remains challenging, because active deflection occurs only at the distal tip and the deflected segment may be too short to reach the calyx.
- Passive (secondary) deflection — the earlier solution: a flexible section just proximal to the point of active deflection. The tip is bent passively off the superior margin of the renal pelvis, moving the point of deflection proximally and effectively lengthening the tip. With passive deflection, the lower pole calyx can be reached in most patients.
- Early 2000s improvement: the deflected segment lengthens as it is deflected against the lower pole infundibulum, giving easier lower pole access and better deflection while working instruments are in place. Most modern scopes incorporate this enhanced primary deflection.
- Failure to reach the lower pole in significant hydronephrosis is due to the limited ability to engage passive secondary deflection — there is no pelvic wall to buttress against.

Calibre, Working Channel and Irrigation

- Tip diameter as small as 4.5 to 7.5 Fr, with a taper improving atraumatic advancement into the ureter.
- All currently available flexible ureteroscopes have a 3.6 Fr working channel.
- Adequate irrigation is possible with working devices up to 3 Fr in situ, but higher-pressure irrigation becomes necessary, particularly when a device occupies the channel.
- Higher-pressure irrigation delivery options: pressurised irrigation bag, roller pump, foot pump, or handheld devices.
- Dual-channel flexible ureteroscope — the Cobra (Richard Wolf): two 3.3 Fr channels permitting unimpeded irrigation through one while a working instrument occupies the other, or both channels used for instruments simultaneously, such as a laser and a basket.

Digital Ureteroscopes

- Digital flexible ureteroscopes are now the preferred upper tract endoscope for most urology services.
- Digital ureteroscopes integrate the camera and light source within the endoscope; no separate camera head is needed because the digital chip — charge-coupled device (CCD) or complementary metal oxide semiconductor (CMOS) — sits in the scope itself.
- Because they dispense with the fiberoptic image bundle, separate light cord and camera head, they may be less prone to damage and longer-lived.
- Trade-off: digital flexible ureteroscopes are generally larger in diameter than the smallest fiberoptic models.
- Consequence of that larger calibre: a prospective comparison found more frequent use of ureteric access sheaths, and more access sheath-related complications, in the digital group.
- Efficacy comparison: stone-free rates statistically equivalent, but operative time significantly shorter with digital scopes — attributed to improved visibility (Fig. 17.10).
- Future direction: further miniaturisation, optimised digital resolution, improved durability.

Single-Use Flexible Ureteroscopes

- LithoVue (Boston Scientific) — the first single-use flexible ureteroscope.
- Rationale: avoiding the high cost of maintaining and repairing reusable endoscopes.
- Durability data on reusable flexible ureteroscopes are variable, but some series report repair needed after only three to five cases, and refurbished scopes appear more prone to damage than new scopes.
- Cost structure of reusables: reprocessing adds roughly \$107 per case in labour and consumables; repair adds approximately \$355 to \$605 per case.
- LithoVue specifications:
 - Light-emitting diode (LED) light source at the tip
 - Digital CMOS sensor, depth of field 2 to 50 mm
 - Tip diameter 7.7 Fr; maximal shaft outer diameter 9.5 Fr
 - Single working channel of 3.6 Fr
 - 270-degree active deflection in both directions
- Cadaveric comparison against reusable fiberoptic and digital ureteroscopes: comparable visibility and manoeuvrability.
- Prospective case-control against a reusable fiberoptic scope: shorter procedure duration and comparable failure rates.
- Economics remain undetermined. Some studies report comparable costs; others report benefit in selected settings — low-volume centres, or larger lower pole stones.
- Competitor single-use flexible ureteroscopes entered the market in the early 2020s, driving costs down: Pusen, YC-FR-A (YouCare), NeoFlex (NeoScope), Axis (Dornier), and the single-use Flex-Xc (Karl Storz).
- Environmental concern about high-volume disposables has been raised, but estimates suggest impacts are comparable to reusables.
- The digital platform opens the way to automated intrarenal pressure and temperature monitoring, and to artificial intelligence applications in future scopes.

Care and Sterilisation

Sites of Damage

- Rigid and flexible ureteroscopes are delicate. Damage to the working channel, deflecting mechanism, or image bundle fibres can render the scope useless.

- One series reported flexible ureteroscope repairs necessary after only 3 to 13 hours of use, usually due to deterioration of the deflecting mechanism.

Working Channel Injury by the Laser Fibre

- The working channel is easily damaged, chiefly during passage of a small 200-micron holmium laser fibre.
- Mechanism, as a sequence: fibre inserted into a scope that is even minimally deflected → fibre tip scrapes the inner working channel → raises a small irregular area → each subsequent pass of a fibre or instrument worsens it → eventual perforation of the channel.
- Once the channel is perforated, sterilisation causes fluid ingress and damage to the imaging system, rendering the scope unusable. This is why perforation is terminal rather than merely a leak.
- Firing the fibre within the working channel also injures the scope; the holmium laser can damage the channel when the fibre tip is very close to the channel's end (Fig. 17.11).
- Preventive rule: keep the fibre tip within the central portion of the field of view when firing.
- The “golden rule” of safe holmium laser lithotripsy: do not step on the pedal if you cannot see the tip of the fibre in contact with the stone. This protects both the ureter and the ureteroscope.
- Also stop firing when repositioning the laser within the scope.

Cleaning and Sterilisation Methods

- After each use: clean the ureteroscope, including the working channel, with warm water and a non-abrasive detergent.
- Sterilisation options, three:
 - Gas — ethylene oxide
 - Soaking in glutaraldehyde solution
 - The Steris system — automated washing and rinsing in peracetic acid solution

Guidewires

Roles and Construction

- Guidewires are essential to urological endoscopy. Uses:
 - Establishing percutaneous or ureteroscopic access
 - Straightening the ureter
 - Guiding dilation of the ureter or percutaneous tract
 - Stent placement
- Wires differ in diameter, rigidity, tip design, materials and coating.
- Selection depends on the task at hand, the patient's anatomy, and the specific upper tract problem.
- Commonest design: a solid stainless steel core with an outer wire wrapped around it.

Materials

- Nitinol (nickel-titanium) inner core: kink-resistant and slightly stiffer.
- Nitinol core with polyurethane outer layer (Zebra, Roadrunner):
 - Less friction of the ureteroscope passing over polyurethane.
 - Stiffer core transmits the operator's “push” more reliably to the ureteroscope tip.
 - The “memory” property of nitinol also permits angling of the wire tip.
- Hydrophilic polymer coating over the polyurethane produces the “glide wire” — exceptionally slippery.
 - Excellent for negotiating impacted ureteric calculi, tortuous ureters and ureteric strictures.

- **But too slippery to serve as a safety wire** — they slide out of the patient too easily.
- **Rule:** when a hydrophilic wire is used for initial access, exchange it for a standard safety wire through an open-ended catheter.
- **Hybrid wires** — hydrophilic tip with a polytetrafluoroethylene (PTFE)-coated shaft — can serve as both access and safety wire in difficult-access cases.

Specifications

- Diameter range 0.018 to 0.038 inches; commonest in use are 0.035 and 0.038 inch.
- Length range 80 to 260 cm; the most useful length for endourology is 145 cm.
- Floppy tip is generally 1 to 3 cm.
- Bentson and Newton designs have flexible tips of up to 15 cm and are seldom used today.
- Movable-core wires allow the core to be partially withdrawn to lengthen the flexible tip.
- Distal tip designs: straight, angled, or “J” tipped.
- Wire rigidity is varied by changing the diameter and design of the inner core wire.
- Stiff wires are useful for straightening tortuous ureters or displacing a sizeable prostatic lobe.
- Despite all advances, a 0.038-inch straight, flexible-tip, PTFE-coated stainless steel wire remains an excellent choice for most cases.

Dilation Devices

Need for Dilation

- Smaller-calibre ureteroscopes have made ureteric dilation less often necessary.
- The need for dilation is directly related to ureteroscope diameter, falling to under 1% for a 7.4 Fr scope.
- Two routes when dilation is needed: passive, by indwelling stent placement, or active, by dilating catheters or balloons.

Specifications

- **Ureteric dilating catheters:** hydrophilic-coated polyurethane, tapered from a 6 Fr tip to a 12 Fr shaft, passed over a wire.
- **Ureteric balloon dilators:** passed over a wire; low profile of 3 to 8 Fr; dilating diameters 12 to 30 Fr; maximum inflation pressures 8 to 20 atmospheres depending on design and balloon material.
- **Practical limit:** dilation beyond 15 Fr is rarely necessary for routine ureteroscopy.
- **Zero-tip ureteric balloon dilators** are useful for dilating immediately adjacent to an impacted ureteric calculus.
- **Ureteroscopic balloon dilators:** 3 Fr, inflatable to 12 Fr, passed directly through the ureteroscope for dilation under direct vision — stenotic infundibula, calyceal diverticular necks.
 - Once inflated, these balloons cannot be withdrawn through the ureteroscope; they must be removed together with the scope.
 - Alternatively used under fluoroscopic guidance, since radio-opaque lines mark the length of the dilating balloon, allowing it to be aligned across the target — taking care not to include a calculus within the balloon length.

Intraluminal Lithotripsy Devices

Electrohydraulic Lithotripsy

- **Mechanism:** shock waves generated by an electric spark at the tip of the electrode.
- Widely available worldwide and cost-effective.

- **Fragmentation is good for most calculi but less effective against dense compositions — cystine and calcium oxalate monohydrate.**
- **Small flexible probes usable with both rigid and flexible ureteroscopes.**
- **Drawbacks: the energy is poorly focused, causing ureteric trauma — this rarely produces stricture but does impair visibility — and it is easy to damage the ureteroscope.**
- **Consequently electrohydraulic lithotripsy is now rarely used, and the AUA recommends against it as a first-line lithotripsy modality in the ureter.**

Pneumatic Lithotripsy

- **Mechanism: mechanical “jackhammer” energy.**
- **Fragmentation is very good and the potential for ureteric injury is low.**
- **Principal drawback: significant stone retropulsion.**
- **Rigid probes are confined to rigid scopes; flexible probes exist but significantly restrict scope deflection.**

Lasers

- **Pulsed dye laser — the first laser successfully used for intraluminal lithotripsy. Abandoned because it could not fragment certain stone compositions and because of high purchase and maintenance costs.**
- **Holmium:YAG — the intraluminal lithotripsy energy of choice since the 1990s.**
 - **Wavelength 2100 nm, absorbed within 4 mm of water — the basis of its safety in urology.**
 - **Mechanism: photothermal reaction with the crystalline matrix of the calculus, producing stone dust rather than fragments, and removing a moderate volume of stone.**
 - **Flexible quartz fibres, usable with both rigid and flexible ureteroscopes, and reusable.**
 - **Effective against every stone composition — the key advantage over electrohydraulic lithotripsy and the pulsed dye laser.**
- **Thulium fibre laser — entered the lithotripsy market in 2020, with several theoretical advantages:**
 - **Wavelength 1940 nm → faster water absorption and a lower penetration depth of 2.5 mm, against 4 mm for holmium.**
 - **Firing frequency up to 2000 Hz, against 120 Hz for holmium.**
 - **Pulse energy as low as 0.025 J — ideal for fine dusting.**
 - **Requires less energy for cooling; therefore no dedicated 240-volt outlet and a smaller footprint than high-powered holmium generators.**
 - **Works with fibres as small as 50 µm, whereas holmium is limited to fibres greater than 200 µm — smaller fibres improve irrigation through small working channels and preserve scope deflection.**
 - **Outstanding concerns: efficacy across all stone types, and thermal injury at high power.**
- **A prospective randomised trial comparing the two laser types showed no appreciable advantage of either in operative time or stone-free rate — theoretical advantage has not yet translated into clinical superiority.**

Stone Retrieval Devices

Construction and Sizing

- **Four essential components: control handle, control wire, sheath, and the device itself.**
- **Shaft diameters range from 1.9 to 7.0 Fr.**
- **Ureteroscopic devices must be 3.0 Fr or smaller, to fit the working channel while leaving room for adequate irrigation.**
- **Larger-diameter devices are for cystoscopic or percutaneous procedures.**

Three-Pronged Grasping Forceps

- The safest instruments for removing calculi with a flexible ureteroscope.
- **Safety mechanism:** the weaker grasp releases the stone if excessive force is applied, thereby preventing ureteric damage. They reliably release stones too large to be withdrawn safely.
- **Critical rule:** avoid basketing an irretrievably large stone when using a flexible ureteroscope without a second working channel.
 - A second working channel permits fragmentation of an unyielding stone trapped in a basket in the other channel.
 - Rigid ureteroscopes with two working channels therefore allow more routine basket use for ureteric calculi.
- Despite being the safest device, three-pronged graspers are seldom used, because basket design and safety have improved.

Stone Baskets: Sheath Materials

- Two sheathing materials in common use: PTFE and polyimide.
- Polyimide — durable but stiff; limits flexible ureteroscope deflection.
- PTFE — limits deflection less, but is less durable and prone to stretching.
- Hybrid design: PTFE at the tip, polyimide in the shaft, combining the advantages of each.

Stone Baskets: Designs

- **Helical baskets:** available in three-wire, four-wire, or double-wire designs with six or more wires.
 - Double-wire designs have improved opening strength, aiding removal of impacted calculi.
 - Some designs unwind the dual wires to expose more wires distally, useful for retrieving multiple small fragments.
 - Helical baskets have round wires and, unlike flat-wire baskets, are safe to rotate within the ureter.
 - Technique, as a sequence: open the basket above the stone → pull down → rotate the basket to engage the stone.
- **Flat-wire baskets:** non-helical, four wires with larger spaces between them to engage larger stones.
 - Originally designed for percutaneous use, where filling the calyx when opened made calyceal stones easier to engage.
 - Also useful for biopsy of papillary ureteric tumours.
- **Tipless grasper-basket hybrids (Grasplit, Dakota, NGage):** designed to combine the safety of three-prong graspers with the secure grasp of a basket. They open at the distal end like a grasper and are advanced onto the stone, permitting easier disengagement of a calculus too large to withdraw safely.

Stone Baskets: Nitinol

- The most recent material innovation is nitinol basket wire.
- **Properties:** soft, with memory; maintains shape; resists kinking. Consequently it opens more safely and disengages stones more reliably than stainless steel.
- Nitinol also permits tipless construction — tipless baskets are soft and safe in the ureter and can be deployed more fully within a calyx without interference from a tip, unlike stainless steel baskets.

Retropulsion Prevention Devices

- Retropulsion is proximal propulsion of the stone during intraluminal lithotripsy.
- **Determinants of the degree of retropulsion:** stone size, stone location, degree of ureteric dilation, and the lithotripsy energy used.

- **Retropulsion into the kidney is not always a problem and is sometimes desirable**, now that renal calculi can be treated with a flexible ureteroscope.
- **Prevention is nonetheless more expeditious**, and is particularly important when no flexible ureteroscope is available.
- **Named devices: Stone Cone and LithoCatch (Boston Scientific), NTrap (Cook Urological), Accordion and CoAx (Accordion Medical).**
- **Stone Cone — 3 Fr**, with a distal coil deployed above the stone before fragmentation. After fragmentation it is **withdrawn to remove fragments**; fragments too large to remove safely are left behind, because the coil simply unravels around the stone.
- **NTrap — 2.8 Fr**, with a mesh-like wire basket deploying perpendicular to the shaft, preventing migration of all but the smallest fragments.

Miscellaneous Devices

- **Cup biopsy forceps — 3 Fr**, for biopsy of sessile tumours.
- **BIGopsy forceps (Cook Urological) — provides a substantially larger specimen**, potentially improving diagnostic yield. Two practical constraints: **it must be backloaded into the ureteroscope**, and after biopsy it must be removed together with the ureteroscope.
- **Electrodes — pencil tip, ball tip, angled and straight tips — used for fulguration and incisional procedures such as endoureterotomy and endopyelotomy.**

Ureteral Access Sheaths

- **Purpose: repeated access to the intrarenal collecting system without replacing the working guidewire on each passage of the endoscope.**
- **Available in outer diameters of 11 to 16 Fr.**
- **Advantages:**
 - **Facilitate retrieval of stone fragments.**
 - **Decrease intrapelvic pressure during ureteroscopy**, which may reduce infectious complications arising from pyelovenous backflow.
- **Disadvantages:**
 - **Size**, with a small potential for ureteric injury.
 - **Most intrarenal stones require only a single passage of the ureteroscope to access and fragment completely — in such cases an access sheath is usually unnecessary.**

Fluoroscopy Equipment

- **Three uses in ureteroscopic procedures: initial ureteric access, monitoring during endoscopy, and stent placement.**
- **Mobile C-arm units are preferable to tables with fixed fluoroscopy units.** Three reasons: **greater mobility**, improved image quality, and less scatter radiation to the surgeon because the x-ray source lies below the patient.
- **Modern C-arms incorporate digital image enhancement and “last image hold” to minimise radiation exposure.**
- **Further dose reduction: image collimation and pulsed fluoroscopy.**
- **The urologist should control the unit by foot pedal**, which speeds the case and minimises excessive fluoroscopy time.
- **“Fluoroless” ureteroscopy series have been published**, calling into question the need for imaging in routine cases.

Ureteroscopy Technique

Preparation for Ureteroscopy

Infection Screening and Prophylaxis

- Urinalysis and urine culture, if indicated, before upper tract endoscopy.
- Urinary tract infections are treated preoperatively with culture-specific antibiotics.
- **Key contrast with cystoscopy:** per the AUA Best Practice Policy Statement, **ALL patients undergoing upper tract endoscopy should receive antibiotic prophylaxis** — unlike diagnostic cystourethroscopy, where prophylaxis is withheld in the absence of risk factors.
- **AUA agents of choice for upper tract endoscopy:** trimethoprim-sulfamethoxazole, or a first- or second-generation cephalosporin.
- **AUA acceptable alternatives:** an aminoglycoside with or without ampicillin, or amoxicillin/clavulanate.
- **EAU position:** the specific antimicrobial should be chosen on the basis of local pathogen prevalence for each type of procedure. Options named by the EAU: trimethoprim with or without sulfamethoxazole, amoxicillin/clavulanate, or a second- or third-generation cephalosporin.

Anaesthesia

- Options: general (endotracheal tube or laryngeal mask), regional, or local with sedation.
- Advantages of general anaesthesia with an endotracheal tube (ET):
 - Renal excursion can be manipulated by adjusting tidal volume and respiratory rate.
 - Periods of apnoea can be induced to facilitate stone treatment.
- Laryngeal mask airway has been associated with significantly greater lasing inefficiency than ET intubation.
- However, meta-analysis comparing neuraxial with general anaesthesia found **no difference in stone-free rate, operative time, pain scores, complication rate, or length of stay**. The anaesthetic choice has not been shown to alter outcomes.

Theatre Readiness

- Endourological procedures should be performed in a fully equipped operating room; the urologist must be prepared for any unanticipated problem.
- Guidewires to have available beyond the conventional PTFE-coated stainless steel wire: angled hydrophilic, nitinol core, and extra-stiff.
- Dilation devices as standard: dilating catheters, high-pressure balloon catheters, zero-tipped balloon catheters.
- Angled catheters that can be reliably rotated or “torqued” assist access beyond impacted calculi, strictures and tortuous ureters.
- The urologist must know the available endoscopes and their working channel sizes, in order to select appropriately sized instruments.
- **Additional flexible and semirigid ureteroscopes should be immediately available** — for unexpected findings or to replace a scope that breaks intraoperatively.

TABLE 17.3 COMMON SUPPLIES FOR URETEROSCOPY

Category	Items
Ureteroscopes, rigid	7 Fr or smaller semirigid; larger scope with straight working channel (optional)
Ureteroscopes, flexible	7.5 Fr; 8.6 Fr or larger; secondary or exaggerated deflection capable
Guidewires (all 0.038 inch)	Angled hydrophilic; straight PTFE-coated; nitinol core with polyurethane coating; extra-stiff

Category	Items
Irrigation	Power irrigation device; high-pressure working port seal
Stone retrieval (≤ 3.0 Fr)	Helical basket; tipless basket; three-pronged grasping forceps or equivalent
Catheters	Dual-lumen; 6–12 Fr dilating; 5 Fr open-ended; 5 Fr angled-tip torqueable
Dilation devices	High-pressure ureteric dilating balloons (5–7 mm); zero-tip dilating balloon
Biopsy devices	3 Fr cup biopsy; flat-wire basket; BIGopsy (optional)
Ureteric stents	5–7 Fr, 20–28 cm, double pigtail
Intraluminal lithotripsy	Holmium laser; pneumatic (optional); electrohydraulic (optional)

- **What the table is for:** it is a readiness checklist, not a list to memorise item by item. The pattern across its rows is **redundancy at every step** — a second scope, a second wire type, a second dilation route, a second energy source.
- **The trap it sets:** reading the 5–7 mm balloon row as the standard ureteric dilation size. It is what is stocked; **the orifice needs only 4 mm.**

Accessing the Ureter

- **Position:** cystolithotomy position.
- **Sequence:** cystoscopy → thorough bladder inspection → placement of a safety guidewire.
- **A simple 0.038-inch flexible-tip PTFE-coated guidewire is usually sufficient as the safety wire.**
- **Purpose of the safety wire:** maintain access and permit stent placement if difficulty arises.
- **The ureter is easily perforated when accessing an impacted stone** — the central hazard of this step.
- **Manipulating the wire past the stone may require an angled hydrophilic wire, an angled torque catheter, or both.**
- **If the wire cannot be safely passed beyond the stone:** inspect the ureter up to the stone with the rigid ureteroscope, which may permit passage of the wire under direct vision.
- **Once access above the stone is achieved,** exchange the hydrophilic wire through a ureteric catheter for a more secure 0.038-inch PTFE-coated wire.
- **Hybrid wires do not require this exchange, serving as both safety and access wire.**
- **If infection above the stone is suspected:** pass an open-ended catheter over the wire and aspirate the renal pelvis.
 - **If the aspirate is turbid → place a stent and postpone ureteroscopy until the infection is treated.** This is the abort trigger.
- **Before proceeding:** drain the bladder, to minimise overdistension during ureteroscopy and reduce buckling of the flexible ureteroscope into the bladder.

Semirigid Ureteroscopy Technique

Division of Labour

- **Flexible ureteroscopes better accommodate the natural tortuosity of the ureter; deflection gives better access to the intrarenal collecting system.**
- **Flexible ureteroscopes are difficult to control in the distal ureter, with poorer irrigation and visualisation.**
- **Therefore semirigid ureteroscopy is generally used for the ureter distal to the iliac vessels.**

Entering the Intramural Ureter

- **Sequence:** cystoscopic placement of a safety guidewire in the ureter → ureteroscope introduced through the urethra into the bladder under direct vision → the contralateral leg may need elevating to permit passage into the intramural ureter.
- **“Riding the rails” technique, as a sequence:** manoeuvre the ureteroscope tip alongside the guidewire posterolaterally → elevate the wire, propping the ureteric orifice open → if needed, pass an additional wire through the ureteroscope into the ureter just beyond the intramural segment → rotate the ureteroscope until it lies directly between the two wires, holding the orifice widely open → advance the scope → remove the additional wire once through the intramural ureter (Fig. 17.12).
- The same technique is useful for negotiating tortuous ureteric segments.

When the Intramural Ureter Will Not Admit the Scope

- **Safest option:** place a ureteric stent for passive dilation and return to theatre in 1 to 2 weeks.
- **Alternative:** balloon dilation of the ureteric orifice — a 4 mm diameter balloon is sufficient.
- **Absolute caution:** do not dilate a segment of ureter containing a stone — this causes ureteric perforation and possible extravasation of the stone.
- **If a stricture beneath a stone prevents safe visualisation and lithotripsy:** use a zero-tip balloon passed over the safety wire immediately below the stone, dilating the stenotic segment while avoiding the stone. The ureteroscope can then be introduced safely and the stone visualised.

The Impacted Stone

- **Preferred manoeuvre:** gently displace the impacted stone proximally with the ureteroscope tip, out of the traumatised segment, for better visibility and safer lithotripsy.
- **If displacement is impossible:** the “drill and core” technique — treat the central portion of the stone with laser energy, using the outer shell to protect the ureteric wall → then safely fragment the collapsed outer shell.
- **As soon as the ureter above the stone becomes visible,** place a safety wire directly through the ureteroscope before completing stone treatment.
- **Two acceptable endpoints for ureteric stones:** “dusting” until no fragment exceeds 2 mm, or cleaving until pieces are small enough for easy basket retrieval.

Flexible Ureteroscopy Technique

Establishing Access

- **Sequence:** guidewire placed in the ureter → cystoscope removed → 10 Fr dual-lumen catheter passed over the wire → gently dilates the ureteric orifice and permits placement of a second, working wire → flexible ureteroscope passed over the taut working wire to the site of pathology.
- Dilation by the dual-lumen catheter is usually sufficient to permit passage of the flexible ureteroscope.
- The working channel of a flexible ureteroscope is not centrally located, so the scope tip sits eccentrically relative to the guidewire.
- **If the scope will not pass the orifice:** rotate it 90 to 180 degrees on the guidewire to reposition the tip relative to the orifice.
- **If difficulty persists:** use a dilating catheter or balloon to widen the orifice.
- **If passage is difficult without significant obstruction:** substitute a nitinol-core, polyurethane-coated guidewire — stiffer and smoother, transmitting the operator’s force more efficiently to the scope tip.

Rates of Formal Ureteric Dilation

- Most ureteroscopy series report formal ureteric dilation needed in roughly 8–25% of patients, a figure that has fallen with smaller-calibre flexible ureteroscopes.
- However, a large multinational registry found dilation still performed in about a third of cases, indicating it remains common practice worldwide.

The Safety Wire

- Traditional rationale: maintain access throughout the procedure, permitting ureteric stent placement should ureteric injury occur.
- Three developments have called the need for a safety wire into question in uncomplicated ureteroscopy: continued miniaturisation of ureteroscopes, widespread adoption of upper tract endoscopic techniques, and the absence of comparative series in the literature.
- The safety wire remains standard; it is its necessity in uncomplicated cases that is disputed, not established as dispensable.

Basic Movements

- Three basic movements: deflection, rotation, and advancing or retreating the scope.
- The reticle marks the plane of deflection; the scope must be rotated to align that plane with the intended target.
- Failure to rotate the ureteroscope adequately is the commonest mistake of the novice ureteroscopist.
- Finest rotational control is obtained by pinching the scope near the meatus, this being closest to the tip.
- Caveat: over-rotation of a fiberoptic scope damages the fibre bundles. This is not an issue with digital scopes.

Irrigation

- Delivery: pressurised irrigation bag, pump, foot pump, or handheld syringe.
- Irrigation must be well controlled and delivered only when needed to clear the visual field.
- Normal saline must be used, to prevent accumulation and absorption of a hypotonic solution and consequent transurethral resection (TUR) syndrome.

Laser Fibre Handling

- The flexible ureteroscope must be straight when the laser fibre is passed, or the working channel will be damaged.
- Sole exception: a ball-tip fibre, which may be passed safely while the scope is actively deflected.
- Fibres of 200 to 365 microns are most commonly used.
- Where significant deflection is required, the 200-micron fibre is preferred — it restricts scope deflection less and permits higher irrigation flow through the shared working channel.

Lower Pole Strategy

- Preference: relocate the stone out of the lower pole rather than rely on a smaller laser fibre.
- A randomised study showed significantly higher stone-free status when lower pole stones were displaced before treatment compared with in situ lithotripsy.
- Not all lower pole stones are accessible in the standard lithotomy position.
- T-tilt position = 15-degree Trendelenburg plus 15-degree airplane rotation away from the side of the operated kidney. It improves lower pole access and stone-free rates relative to lithotomy.
- When relocation is impossible: prolonged manual deflection while treating lower pole stones in situ is a critical factor degrading the durability of reusable ureteroscopes. Therefore, if economically feasible,

use a disposable ureteroscope for large lower pole stones that cannot be relocated, given the high likelihood of scope damage.

Protecting the Scope from the Laser

- The laser is a common cause of ureteroscope damage.
- Rule one: do not activate the laser unless the fibre tip is seen in contact with the stone.
- Rule two: do not activate the laser if the aiming beam is not visible — absence of the aiming beam may indicate fibre damage, and firing through a broken fibre causes significant scope injury.

Postoperative Analgesia

- Nonsteroidal anti-inflammatory drugs (NSAIDs), an alpha-adrenergic blocker, or both.

KEY POINTS: URETEROSCOPY

- Semirigid ureteroscopy is used below the iliac vessels; flexible ureteroscopy above.
- Semirigid ureteroscopes with two working channels permit better irrigation and allow a lithotripsy device to be passed through one channel to fragment a stone engaged in a basket in the other.
- Flexible ureteroscopes must be straight when passing a laser fibre, or the working channel will be damaged.
- The golden rule of safe laser lithotripsy: do not step on the pedal if you cannot see the tip of the fibre in contact with the stone.
- Nitinol stone baskets maintain their shape, resist kinking, and disengage stones more reliably than stainless steel baskets.
- Ureteral access sheaths facilitate repeated passage of flexible ureteroscopes and decrease intrapelvic pressure.
- Mobile C-arm fluoroscopy is preferred for greater mobility, improved image quality, and less scatter radiation to the surgeon than fixed-unit urology tables.
- A routine preoperative antibiotic should be given to all patients undergoing ureteroscopy.
- Normal saline should be used as irrigation during ureteroscopy to prevent absorption of a hypotonic solution.

CONCLUSIONS

- Over the past 150 years, advances in endoscope design have transformed urological surgery.
- Visualisation of both upper and lower urinary tracts is now routine with rigid and flexible endoscopes.
- Office flexible cystoscopy allows efficient evaluation of a range of conditions with minimal discomfort.
- Rigid cystourethroscopy in theatre manages a wide array of benign and malignant bladder and urethral pathology with limited morbidity.
- Improvements in flexible ureteroscopes, working instruments, energy delivery and technique have improved treatment of upper tract disease.
- The role of ureteroscopy in complex intrarenal calculi, ureteric obstruction and upper tract tumours will continue to grow.